

A Practical Guide to RMSprop optimizer

TOPIC -- RMSPROP OPTIMIZER

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Further reading Bottou, Léon (2004), "Stochastic Learning", Advanced Lectures on Machine Learning, LNAI, vol. 3176, Springer, pp. 146–168, ISBN 978-3-540-23122-6 Buduma, Nikhil; Locascio, Nicholas (2017), "Beyond Gradient Descent", Fundamentals of Deep Learning : Designing Next-Generation Machine Intelligence Algorithms, O'Reilly, ISBN 9781491925584 LeCun, Yann A.; Bottou, Léon; Orr, Genevieve B.; Müller, Klaus-Robert (2012), "Efficient BackProp", Neural Networks: Tricks of the Trade, Springer, pp. 9–48, ISBN 978-3-642-35288-1 Spall, James C. (2003), Introduction to Stochastic Search and Optimization, Wiley, ISBN 978-0-471-33052-3

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AdaGrad AdaGrad (for adaptive gradient algorithm) is a modified stochastic gradient descent algorithm with per-parameter learning rate, first published in 2011. Informally, this increases the learning rate for sparser parameters and decreases the learning rate for ones that are less sparse. This strategy often improves convergence performance over standard stochastic gradient descent in settings where data is sparse and sparse parameters are more informative. Examples of such applications include natural language processing and image recognition. It still has a base learning rate η , but this is multiplied with the elements of a vector $\{G_j, j\}$ which is the diagonal of the outer product matrix

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This procedure will remain numerically stable virtually for all η $\{\displaystyle \eta\}$ as the learning rate is now normalized. Such comparison between classical and implicit stochastic gradient descent in the least squares problem is very similar to the comparison between least mean squares (LMS) and normalized least mean squares filter (NLMS). Even though a closed-form solution for ISGD is only possible in least squares, the procedure can be efficiently implemented in a wide range of models. Specifically, suppose that $Q_i(w)$ $\{\displaystyle Q_{\{i\}}(w)\}$ depends on w $\{\displaystyle w\}$ only through a linear combination with features x_i $\{\displaystyle x_{\{i\}}\}$, so that we can write $\nabla_w Q_i(w) = -q(x_i'w)x_i$ $\{\displaystyle \nabla_{\{w\}}Q_{\{i\}}(w)=-q(x_{\{i\}}'w)x_{\{i\}}\}$, where $q(\cdot) \in \mathbb{R}$ $\{\displaystyle q(\cdot) \in \mathbb{R}\}$ may depend on x_i, y_i $\{\displaystyle x_{\{i\}}, y_{\{i\}}\}$ as well but

not on w except through $x_i' w$. Least squares obeys this rule, and so does logistic regression, and most generalized linear models. For instance, in least squares, $q(x_i' w) = y_i - x_i' w$, and in logistic regression $q(x_i' w) = y_i - S(x_i' w)$, where $S(u) = e^u / (1 + e^u)$ is the logistic function. In Poisson regression, $q(x_i' w) = y_i - e^{x_i' w}$, and so on. In such settings, ISGD is simply implemented as follows. Let $f(\xi) = \eta q(x_i' w_{\text{old}}) + \xi \|x_i\|^2$, where ξ is scalar. Then, ISGD is equivalent to:

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$$w^{\text{new}} = w^{\text{old}} + \frac{\eta}{1 + \eta \|x_i\|^2} (y_i - x_i' w^{\text{old}}) x_i.$$

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where, γ is the forgetting factor. The concept of storing the historical gradient as sum of squares is borrowed from Adagrad, but "forgetting" is introduced to solve Adagrad's diminishing learning rates in non-convex problems by gradually decreasing the influence of old data. And the parameters are updated as,

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$$w(t) := w(t-1) - \eta \frac{w(t) v(t)}{\sqrt{\hat{v}(t)}} + \varepsilon$$

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RMSProp has shown good adaptation of learning rate in different applications. RMSProp can be seen as a generalization of Rprop and is capable to work with mini-batches as well opposed to only full-batches.

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Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

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$$w^{\text{new}} := \arg \min_w \{ Q_i(w) + \frac{1}{2} \eta \|w - w^{\text{old}}\|^2 \} . \quad \{\displaystyle w^{\text{new}} := \arg \min_w \left\{ Q_i(w) + \frac{1}{2} \eta \|w - w^{\text{old}}\|^2 \right\} .\}$$

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A compromise between computing the true gradient and the gradient at a single sample is to compute the gradient against more than one training sample (called a "mini-batch") at each step. This can perform significantly better than "true" stochastic gradient descent described, because the code can make use of vectorization libraries rather than computing each step separately as was first shown in where it was called "the bunch-mode back-propagation algorithm". It may also result in smoother convergence, as the gradient computed at each step is averaged over more training samples. The convergence of stochastic gradient descent has been analyzed using the theories of convex minimization and of stochastic approximation. Briefly, when the learning rates η $\{\displaystyle \eta\}$ decrease with an appropriate rate, and subject to relatively mild assumptions, stochastic gradient descent converges almost surely to a global minimum when the objective function is convex or pseudoconvex, and otherwise converges almost surely to a local minimum. This is in fact a consequence of the Robbins–Siegmund theorem.

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Adam (short for Adaptive Moment Estimation) is a 2014 update to the RMSProp optimizer combining it with the main feature of the Momentum method. In this optimization algorithm, running averages with exponential forgetting of both the gradients and the second moments of the gradients are used. Given parameters $w(t)$ $\{\displaystyle w^{(t)}\}$ and a loss function $L(t)$ $\{\displaystyle L^{(t)}\}$, where t $\{\displaystyle t\}$ indexes the current training iteration (indexed at 1 $\{\displaystyle 1\}$), Adam's parameter update is given by: